Technical Note

Power Electronics and Control Systems

AFE control in the uvw frame

the five MATLAB Function blocks of `afe_uvw_ctrl`: sequences, fault ride-through, reactive current loops, per-phase resonant PI and islanding detection

The controller of the model ***`afe_uvw_three_phase`*** *as it is written in the library package* ***`masked_models/afe_uvw_ctrl`*** *of* ***`pwr-control/library`****: the algorithm of each block, equation by equation, with the discrete implementation of the resonant PI current controller and what it means for the poles of the loop.*

Authors:	**Revision:**	**Date:**
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Abstract: The control of the three-phase active front end (AFE) of the model `afe_uvw_three_phase` works in the stationary frame: one resonant PI per phase makes the converter current follow a sinusoidal reference, while the slow loops (reactive current of the positive and the negative sequence, fault ride-through) work on moving averages of the grid quantities in the positive and the negative synchronous frame. The library package `afe_uvw_ctrl` rewrites this control as five MATLAB Function blocks with one parameter structure. This note describes the algorithms as they are in the code: the Clarke and sequence transformations and the moving average with a fractional window, the voltage-rate observers and the state machine of the fault ride-through with the reactive current references of VDE-AR-N 4110, the three clamped PIs of the reactive current loops, the per-phase resonant PI with its forward-backward Euler discretization, the vector limiter, the output delay and the zero-sequence loop, and the new passive islanding detector. The resonant PI is analysed in detail: the forward Euler resonator of the model has its poles just outside the unit circle with the damping of the model ($\delta = 0.015$, $|z| = 1.000182$), so that controller alone is marginally unstable and its gain at 50 Hz is 11.4 instead of the 4.3 of the continuous design, while the current loop with the plant is stable (phase margin 69°, sensitivity -41 dB at 50 Hz). Since rev. 03 of the library the block uses the updated x_1 in the update of x_2 (forward-backward Euler), whose poles are inside the circle for any $\delta > 0$, with $\delta = 0.005$ to keep the gain at the grid frequency and the same loop (sensitivity -41 dB). The parameter structure, the mapping to the model init, the checks against the original block diagrams and the open points close the note.



Revision history

Rev.	Date	Description	Authors
00	November 2026	First issue: the five blocks of `afe_uvw_ctrl` (library PR #35), resonant PI analysis, parameters, checks, open points.	Davide Bagnara
01	November 2026	The forward-backward Euler resonator proposed in rev. 00 is now in the block (library rev. 03) with $\delta = 0.005$: Section 6.3 describes it, Section 6.4 keeps the analysis of the forward Euler resonator of the model; figures, loop table, parameter table, differences and open points updated.	Davide Bagnara

Project Repository: library.git (masked_models/afe_uvw_ctrl)

Framework: MATLAB/Simulink, GNU Octave, Python



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Nomenclature

Symbol	Meaning
AFE	active front end: three-phase PWM converter between a DC link and the grid
uvw, abc	the three phases; the model calls the stationary per-phase frame “uvw”, the code uses the indices a, b, c
$\alpha\beta$	stationary two-axis frame (amplitude-invariant Clarke transform)
$\xi\eta$	rotating two-axis frame; the “positive frame” turns by $+\theta$, the “negative frame” by $-\theta$
pos, neg	positive and negative sequence of a three-phase quantity
pu	per unit: a quantity divided by its base (peak phase voltage, peak phase current of the converter)
θ, ω_{pu}	grid angle and grid pulsation (in pu of ω_{nom}) given by the PLL of the model
ω_{nom}, f_0	nominal grid pulsation $2\pi f_0$, $f_0 = 50\,\mathrm{Hz}$
t_{s}	control period, 125 µs in the model (4 kHz PWM, double update)
u_{g}, i_{g}, i_{s}	grid voltage, grid current and stack (converter) current, in pu
$u_{\mathrm{g,abc}}$, $u_{\mathrm{g},\alpha\beta}$	the same quantity in phase components and in $\alpha\beta$ components
k_{p}, k_{i}, δ	proportional gain, resonant gain and damping of the resonant PI
x_1, x_2	states of the resonator of one phase; x_2 is its output
z	variable of the z transform; $z = e^{j\omega t_{\mathrm{s}}}$ on the unit circle
$C(s)$, $C_{\mathrm{d}}(z)$	continuous and discrete transfer function of the resonant PI
$P(z)$, $L(z)$, $S(z)$	plant, loop gain and sensitivity $1/(1+L)$ of the per-phase current loop
FRT, LVRT	fault ride-through, low-voltage ride-through
ROCOF	rate of change of frequency, in Hz/s
PI, PR	proportional-integral, proportional-resonant controller
N, n, φ	length of the averaging window in samples (real), its integer part and its fractional part
sat	saturation (clamp) of a value between −limit and +limit

1 Introduction

The model `afe_uvw_three_phase` of `pwr-control/modelization-and-control` (file `afe_rst_ctrl.slx`) is a three-phase AFE whose current control works per phase, in the stationary frame, with resonant PI controllers. Its control subsystems were built from Simulink blocks and C-Caller blocks. The library package `masked_models/afe_uvw_ctrl` of `pwr-control/library` [1] rewrites them as five MATLAB Function blocks (Figure 1, Table 1), one function each, all reading the same parameter structure `p`:

- `grid_voltage_sequences`: positive and negative sequence of the grid voltage, averaged;
- `frt_vde4110`: detection of a grid fault and reactive current references during the fault;

- `grid_current_sequences_q_ctrl`: sequences of the grid current and the three slow loops that set the reactive components of the converter current;
- `current_ctrl_uvw`: the fast per-phase current control with the resonant PIs, the vector limiter and the zero-sequence loop;
- `islanding_detector`: a new passive islanding detector (the model only measured the 7th harmonic).

This note describes the algorithms as they are written in the code (folder `src/` of the package). It does not describe the Simulink diagrams that were replaced: Section 10 lists where the code differs from them. The equations use the same names as the code, so that the reader can go from the note to the function and back. Section 6 on the resonant PI is the longest, because the discretization of the resonator has a property that is easy to miss: with the forward Euler step of the model the resonator alone is marginally unstable, and the loop is stable only together with the plant. The block uses a different step since rev. 03 of the library, with its damping chosen to keep the same loop.

Three things are outside the five blocks and stay in the model: the PLL that gives the grid angle θ and the pulsation ω_{pu}, the DC-link voltage controller that gives the active current reference $i_{\xi,\mathrm{ref}}$ (and reads the power-limitation flag of the FRT), and the modulator that turns the voltage reference $u_{\alpha\beta\gamma}$ into the gate signals.

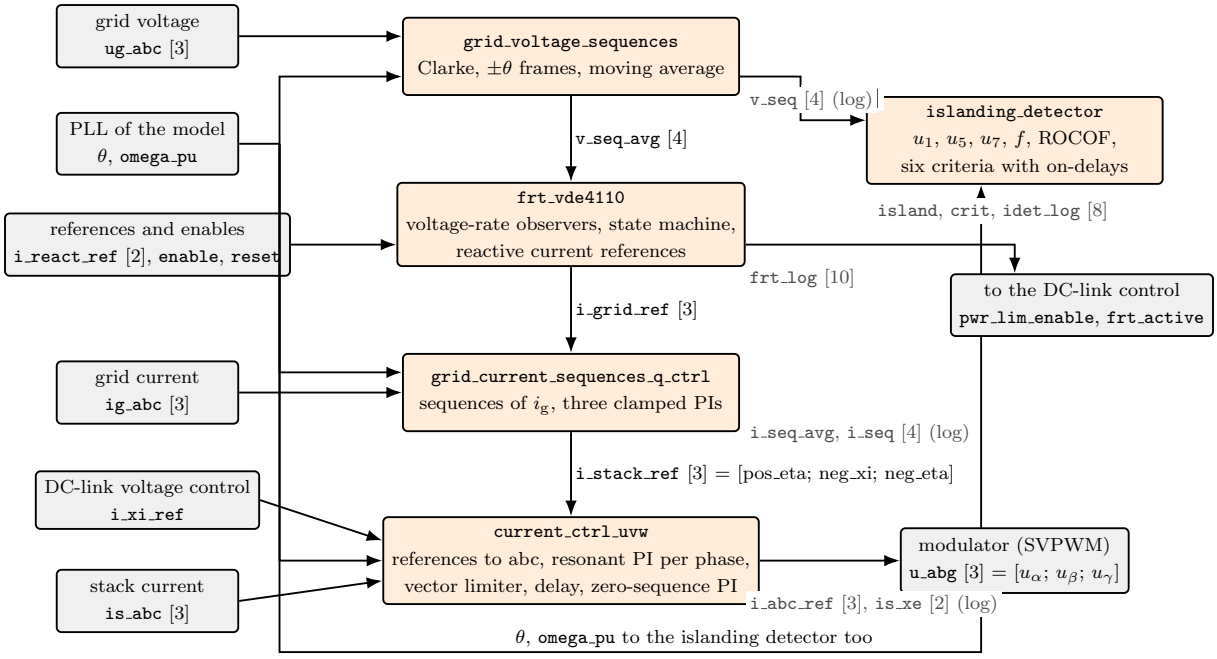


Figure 1: The five blocks of `afe_uvw_ctrl` and the signals between them (sizes in brackets). Grey: what stays in the model (measurements, PLL, DC-link control, modulator). All blocks run once per control period t_{s} inside the triggered control subsystem of the model.



Table 1: Ports of the five blocks (column vectors, pu unless noted). Every block also has the mask parameter `p`.

Block	Inputs	Outputs
`grid_voltage_sequences`	`ug_abc` [3], `theta` [rad], `omega_pu`	`v_seq_avg` [4], `v_seq` [4] (both [pos_xi; pos_eta; neg_xi; neg_eta]), `ug_ab` [2]
`frt_vde4110`	`v_seq_avg` [4], `i_react_ref` [2] ([pos; neg]), `enable`, `reset`	`i_grid_ref` [3], `frt_active`, `pwr_lim_enable`, `frt_log` [10]
`grid_current_sequences_q_ctrl`	`ig_abc` [3], `theta`, `omega_pu`, `i_grid_ref` [3]	`i_stack_ref` [3], `i_seq_avg` [4], `i_seq` [4]
`current_ctrl_uvw`	`i_xi_ref`, `i_stack_ref` [3], `is_abc` [3], `theta`	`u_abg` [3], `i_abc_ref` [3], `is_xe` [2]
`islanding_detector`	`ug_ab` [2], `theta`, `omega_pu`, `enable`	`island`, `crit` (bit mask), `idet_log` [8]

Discrete time. The blocks have no sample time of their own: they inherit it and integrate with `p.ts`, so they must run exactly once per control period. In the model this is the triggered subsystem `afe_control`, clocked by the PWM carrier with double update ($t_\mathrm{s} = 1/(2f_\mathrm{pwm}) = 125\,\mu\mathrm{s}$). The states of every block live in `persistent` variables and start from zero at the start of the simulation; the start-up times of the blocks (`t_enable`, `t_obs_release`, `t_avg_start`) count from the first call.

2 Frames and the moving average

Three operations appear in several blocks. They are described once here.

2.1 Clarke transformation

The three phase quantities become two stationary components with the amplitude-invariant Clarke transformation (the zero sequence is dropped):

$$\alpha = \tfrac{2}{3}\left(x_\mathrm{a} - \tfrac{1}{2}x_\mathrm{b} - \tfrac{1}{2}x_\mathrm{c}\right), \qquad \beta = \tfrac{1}{\sqrt{3}}\left(x_\mathrm{b} - x_\mathrm{c}\right). \tag{2.1}$$

A balanced three-phase set of amplitude 1 gives a vector of length 1 in $\alpha\beta$, so the pu values do not change. The inverse, used for the current references and for the voltage of the plant, is $x_\mathrm{a} = \alpha$, $x_\mathrm{b} = -\alpha/2 + \sqrt{3}\beta/2$, $x_\mathrm{c} = -\alpha/2 - \sqrt{3}\beta/2$.

2.2 Positive and negative sequence frames

With $c = \cos\theta$ and $s = \sin\theta$, θ the angle of the PLL, the $\alpha\beta$ vector is turned into the *positive frame* (which rotates with the positive sequence) and into the *negative frame* (which rotates the other way):

$$\begin{aligned} \text{positive:} \quad & \xi_\mathrm{pos} = \alpha c + \beta s, \quad \eta_\mathrm{pos} = -\alpha s + \beta c, \\ \text{negative:} \quad & \xi_\mathrm{neg} = \alpha c - \beta s, \quad \eta_\mathrm{neg} = \alpha s + \beta c. \end{aligned} \tag{2.2}$$



The positive frame is the usual dq (Park) frame: ξ is d, η is q. When the PLL is locked to the positive sequence of the grid voltage, that voltage is a constant vector [1; 0] in the positive frame. A negative-sequence component (an unbalance) is constant in the negative frame and rotates at 2ω in the positive frame, and the other way round. The 5th harmonic of the grid voltage is a negative sequence and appears at 6ω in both frames; the 7th is a positive sequence and appears at 6ω too. So the four components of (2.2) are a DC value (the sequence that belongs to the frame) plus ripple at even multiples of ω.

2.3 Moving average with a fractional window

The ripple is removed by a moving average whose window T follows the measured grid frequency:

$$T = \frac{2\pi}{\omega_{\text{nom}}\, k_{\text{avg}}\, w}, \qquad w = \text{sat}\big(|\omega_{\text{pu}}|,\ \omega_{\text{pu,min}},\ \omega_{\text{pu,max}}\big), \tag{2.3}$$

with k_{avg} = `avg_order`. The model uses $k_{\text{avg}} = 2$, half a grid period (10 ms at 50 Hz): the average over half a period cancels every component at an even multiple of ω, so the 2ω term of the other sequence and the 6ω terms of the 5th and 7th harmonics disappear together. The window is $N = T/t_{\text{s}}$ samples long, and N is not an integer in general (80 samples at exactly 50 Hz, 79.68 at 50.2 Hz). The code keeps a circular buffer of the last samples and computes

$$y(k) = \frac{1}{n+\varphi}\left(\sum_{j=0}^{n-1} u(k-j) + \varphi\, u(k-n)\right), \qquad n = \lfloor N \rfloor, \quad \varphi = N - n, \tag{2.4}$$

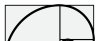
so the next older sample enters with the fractional weight φ and the window length changes without jumps when the frequency moves. While the buffer fills up (fewer samples than N), the mean of the samples received so far is used, which is unbiased from the first sample. Before `t_enable` (25 ms in the model) the averages are forced to zero, as the `dmavg_flt` blocks of the model did. The delay of a moving average is half its window: 5 ms for the half-period window, which is the delay that the slow loops of Section 5 see. The same function (`moving_average_frac`) sits at the end of the three files that use it, with a buffer of 4096 samples, enough for a full period down to $\omega_{\text{pu,min}}$.

3 Grid voltage sequences (grid_voltage_sequences)

Inputs: `ug_abc`, `theta`, `omega_pu`. Three steps:

1. Clarke (2.1) of the grid voltage, output as `ug_ab` (used by the islanding detector);
2. rotation (2.2) into the two frames, output as `v_seq` = [pos_xi; pos_eta; neg_xi; neg_eta];
3. moving average (2.4) of the four components with the window (2.3) of `p.vseq`, output as `v_seq_avg`.

With a locked PLL, `v_seq_avg` is [1; 0; 0; 0] on a clean, balanced grid. During a fault, pos_xi drops to the retained voltage and neg_xi, neg_eta give the unbalance; their length $|v_{\text{neg}}| = \sqrt{\text{neg_xi}^2 + \text{neg_eta}^2}$ is what the FRT block compares with its thresholds. The block has no gains: only the window order, the limits of the measured pulsation and the enable time (Table 3).



4 Fault ride-through (frt_vde4110)

Inputs: v_seq_avg, the reactive current references of normal operation i_react_ref = [pos; neg], enable and reset. The block has two parts: the detection of the fault (the LVRT_detector of the model and its two voltage-rate observers) and the current references during the fault (FRT-PosNegReactiveCurrentReferences of the model).

4.1 Voltage-rate observers

The detector needs the rate of change of the voltage, which cannot be read from a difference of two samples because of the ripple. Two double-integrator observers, one on pos_xi and one on neg_xi, estimate it. For an input u, with the states x (the filtered value) and v (its rate), the update of one step is

$$e = u - x, \qquad x^{+} = x + t_{\mathrm{s}}\,(v + k_x e)\,, \qquad v^{+} = v + t_{\mathrm{s}}\,k_v\,e. \tag{4.1}$$

This is the forward Euler discretization of the continuous observer $\dot{x} = v + k_x(u-x)$, $\dot{v} = k_v(u-x)$, whose poles are the roots of $s^2 + k_x s + k_v$. The model places the poles at $[-0.1,\ -0.4]\,\omega_{\mathrm{nom}}$ (time constants 32 ms and 8 ms) with acker on the discrete pair $A = [1\ t_{\mathrm{s}};\ 0\ 1]$, $C = [1\ 0]$; the library file afe_uvw_ctrl_param_example.m computes the same gains in closed form,

$$z_{1,2} = e^{p_{1,2} t_{\mathrm{s}}}, \qquad l_1 = 2 - (z_1 + z_2), \qquad l_2 = \frac{z_1 z_2 - 1 + l_1}{t_{\mathrm{s}}}, \qquad k_x = \frac{l_1}{t_{\mathrm{s}}}, \quad k_v = \frac{l_2}{t_{\mathrm{s}}}, \tag{4.2}$$

which gives $k_x = 156.0/\mathrm{s}$ and $k_v = 3909/\mathrm{s}^2$ (the continuous-time values would be $-(p_1 + p_2) = 157.1$ and $p_1 p_2 = 3948$). The observers are held at zero before t_obs_release (25 ms), so that the start-up of the sequence averages, which jump from zero to one, is not read as a voltage rate. The detector uses $|v|$ of each observer, called vxi_p_dot and vxi_n_dot (both in the log).

4.2 Fault detection state machine

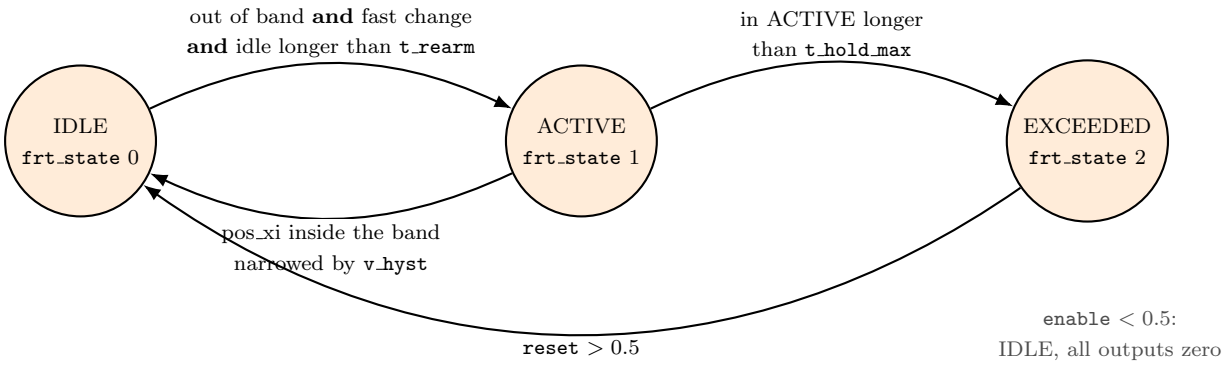


Figure 2: State machine of the fault detection. Out of band: pos_xi > v_pos_high or pos_xi < v_pos_low or $|v_{\mathrm{neg}}|$ > v_neg_max; fast change: vxi_p_dot > v_pos_dot_max or vxi_n_dot > v_neg_dot_max.



The state machine (Figure 2) runs when `enable` is high. From IDLE it goes to ACTIVE when the positive-sequence voltage leaves the band [0.85, 1.15] pu or the negative sequence exceeds 0.1 pu, *and* one of the two voltage rates is above 2 pu/s, *and* the detector has been idle for more than `t_rearm` (125 ms). The rate condition makes the detector react to a fault and not to a slow drift of the voltage. It goes back to IDLE when pos_xi is inside the band narrowed by `v_hyst` = 0.1 pu on both sides ($0.95 \le$ pos_xi < 1.05), and to EXCEEDED when it has been ACTIVE for more than `t_hold_max` (7.625 s, the fault lasted too long); EXCEEDED is left only with `reset`. The output `frt_active` is 1 in ACTIVE; `frt_log(1)` gives the state after the update of the step.

Two more flags are set in ACTIVE. The *power limitation* `pwr_lim_enable`, read by the DC-link controller of the model, is set when pos_xi $<$ `pwrlim_v_pos` (0.8 pu) or $|v_{\mathrm{neg}}| >$ `pwrlim_v_neg` (0.1 pu), and released with hysteresis when pos_xi $>$ `pwrlim_v_pos` + `pwrlim_hyst` (0.9) and $|v_{\mathrm{neg}}| <$ `pwrlim_v_neg_release` (0.7, the value of the model's diagram: see the open points). The *reduced dynamic support* `red_dyn_active` is set when `reddyn_enabled` is 1 and pos_xi $<$ `reddyn_v_pos` (0.7) or $|v_{\mathrm{neg}}| >$ `reddyn_v_neg` (0.2); it switches the current limits to the reduced set. The model keeps it off (constant 0); in the block it is a parameter.

4.3 Pre-fault voltages

The reactive current of VDE-AR-N 4110 [3] is proportional to the *change* of the voltage with respect to the value before the fault. The block keeps a moving average of `v_seq_avg` over `t_avg_prefault` (0.5 s, a running sum in a circular buffer of 8192 samples) that is *frozen* while `frt_active` is 1, so that during the fault it still holds the pre-fault values, and that starts only at `t_avg_start` (50 ms), after the start-up of the sequence averages, so that their initial zeros do not enter the window. The difference $\Delta v =$ `v_seq_avg` $-$ `v_prefault` is the voltage change; its four components are in the log.

4.4 Reactive current references

With $k =$ `k_frt_ref` (2 in the model, the gain of VDE-AR-N 4110), during the fault

$$\begin{aligned} i_{\mathrm{pos},\eta} &= i_{\mathrm{react,ref}}(1) + k\,\Delta v_{\mathrm{pos},\xi}, \\ i_{\mathrm{neg},\eta} &= i_{\mathrm{react,ref}}(2) - k\,\Delta v_{\mathrm{neg},\xi}, \\ i_{\mathrm{neg},\xi} &= \qquad\qquad + k\,\Delta v_{\mathrm{neg},\eta}, \end{aligned} \tag{4.3}$$

and outside the fault the first two are the inputs and the third is zero. The positive-sequence reference moves along η (the axis in quadrature with the voltage: a reactive current) by k times the drop of the voltage; a dip to 0.55 pu with $k = 2$ gives -0.9 pu on top of the normal reference. The negative-sequence reference is the negative-sequence voltage change turned by 90° in the negative frame ($\Delta v_\xi \to -i_\eta$, $\Delta v_\eta \to +i_\xi$): a reactive current against the unbalance. The direction of the support in the real circuit depends on the direction in which the model measures the grid current; this is discussed in Section 11.

Each of the three references is limited to $\pm$`lim_pos_eta`, `lim_neg_xi`, `lim_neg_eta` (1.0, 0.5,



0.5 pu), or to the reduced limits (0.1 pu each) when `red_dyn_active` is 1, and then smoothed by a first-order low pass at `f_lpf_ref` = 161 Hz, written as in the `u_161Hz_flt` blocks of the model:

$$y^{+} = g_0\, u + (1 - g_0)\, y, \qquad g_0 = 2\pi f_{\mathrm{lpf}}\, t_{\mathrm{s}} = 0.126. \tag{4.4}$$

The output `i_grid_ref` = [pos_eta; neg_xi; neg_eta] is the reference of the grid current for the slow loops of the next block.

5 Grid current sequences and reactive current control (grid_current_sequences_q_ctrl)

Inputs: `ig_abc`, `theta`, `omega_pu`, `i_grid_ref`. The first part repeats Section 3 on the grid current: Clarke, the two frames, the moving average with the window of `p.iseq`; the outputs are `i_seq` and `i_seq_avg` = [pos_xi; pos_eta; neg_xi; neg_eta].

The second part is three PI controllers that make the averaged grid current follow `i_grid_ref` by moving the references of the stack current, which the inner loop of Section 6 then tracks. The positive-sequence active current (ξ) is not controlled here: its reference comes from the DC-link voltage controller. Each PI works on

$$e = s_{\mathrm{err}}\,\big(\text{reference} - \text{measured}\big), \qquad s_{\mathrm{err}} = \texttt{err_sign} = -1, \tag{5.1}$$

because the model measures the grid current *towards* the converter and the stack current *out of* the converter: a grid current above its reference asks for *more* stack current. In the model this is a Sum block (reference − measured) followed by a gain −1; in a model where the two currents are measured in the same direction, `err_sign` = +1.

The PI is the function `pi_clamped` at the end of the file,

$$u_{\mathrm{p}} = k_{\mathrm{p}}\, e, \qquad u_{\mathrm{i}} \leftarrow u_{\mathrm{i}} + k_{\mathrm{i}}\, t_{\mathrm{s}}\, e, \qquad u = \mathrm{sat}(u_{\mathrm{p}} + u_{\mathrm{i}},\ \pm u_{\mathrm{lim}}), \qquad u_{\mathrm{i}} \leftarrow u - u_{\mathrm{p}}, \tag{5.2}$$

backward Euler on the integral (the present error is included) and the integral state clamped to what the limit leaves to it: when the output is limited, u_{i} is set to $u_{\mathrm{lim}} - u_{\mathrm{p}}$, so the integrator never winds up beyond the limit and leaves the limit as soon as the error changes sign. This is the anti-windup of the Simulink PI of the model (a unit delay fed with saturation minus the proportional part). Gains and limits: `kp_pos`, `ki_pos`, `lim_pos_eta` for the positive-sequence η loop (0.25, 35/s, 1.0 pu); `kp_neg`, `ki_neg` for the two negative-sequence loops (0.25, 35/s) with the limits `lim_neg_xi` (1.0 pu, the model uses the positive limit here too) and `lim_neg_eta` (0.5 pu). The output is `i_stack_ref` = [pos_eta; neg_xi; neg_eta].

These loops see the plant "inner current loop plus the half-period moving average": a delay of about 5 ms plus the lag of the current loop. With $k_{\mathrm{i}} = 35$/s and $k_{\mathrm{p}} = 0.25$ the loops are slow (the integral alone crosses at 5.6 Hz), which is what the averaged measurement allows. In the test bench (Section 9) the averaged grid current reaches a reference step within 5×10^{-5} pu in steady state.

6 Stack current control with per-phase resonant PI (current_ctrl_uvw)

Inputs: `i_xi_ref` (positive-sequence active current, from the DC-link controller), `i_stack_ref` = [pos_eta; neg_xi; neg_eta], the measured stack currents `is_abc` and `theta`. Figure 3 shows the four steps of the block for one phase.

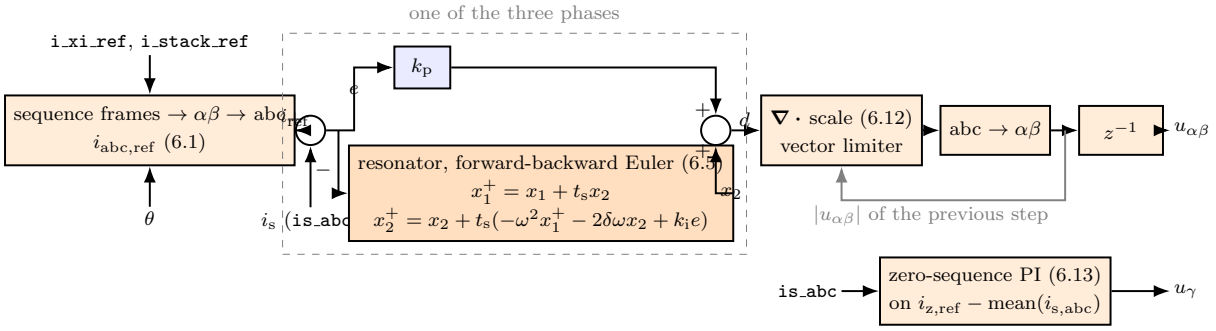


Figure 3: `current_ctrl_uvw`: references, per-phase resonant PI (three copies, one per phase, states in the 2×3 matrix `x_res`), vector limiter on the magnitude of the previous $\alpha\beta$ output, transformation to $\alpha\beta$, one-step delay, and the zero-sequence PI that gives u_γ.

6.1 Current references in phase components

The four sequence references are turned back into the stationary frame: the positive pair by $+\theta$, the negative pair by $-\theta$, summed in $\alpha\beta$ and transformed to abc with the inverse Clarke transformation,

$$\begin{aligned} \alpha_{\text{ref}} &= \big(i_{\text{pos},\xi} c - i_{\text{pos},\eta} s\big) + \big(i_{\text{neg},\xi} c + i_{\text{neg},\eta} s\big), \\ \beta_{\text{ref}} &= \big(i_{\text{pos},\xi} s + i_{\text{pos},\eta} c\big) + \big(-i_{\text{neg},\xi} s + i_{\text{neg},\eta} c\big), \end{aligned} \qquad i_{\text{abc,ref}} = \begin{bmatrix} \alpha_{\text{ref}} \\ -\frac{1}{2}\alpha_{\text{ref}} + \frac{\sqrt{3}}{2}\beta_{\text{ref}} \\ -\frac{1}{2}\alpha_{\text{ref}} - \frac{\sqrt{3}}{2}\beta_{\text{ref}} \end{bmatrix}, \tag{6.1}$$

with $i_{\text{pos},\xi}$ = `i_xi_ref` and the other three from `i_stack_ref`. The references are sinusoids at the grid frequency, with a negative-sequence part when the slow loops ask for one. They are output as `i_abc_ref` for the log. The error of each phase is $e = i_{\text{abc,ref}} - i_{\text{s,abc}}$.

6.2 The resonant PI in continuous time

A PI controller cannot track a sinusoidal reference without error, because its gain at the grid frequency is finite. The resonant PI (also called PR, proportional-resonant [5]) replaces the integrator by a resonator tuned at the grid pulsation ω,

$$C(s) = k_{\text{p}} + k_{\text{i}} \frac{s}{s^2 + 2\delta\omega s + \omega^2}, \tag{6.2}$$

which has the same role for a sinusoid that the integrator has for a constant (internal model principle). With $\delta = 0$ the gain at ω is infinite and the steady-state error at the grid frequency



is zero; the class of the model has $\delta = 0.015$ (`delta_res`), which gives a finite peak

$$|C(j\omega)| = k_\mathrm{p} + \frac{k_\mathrm{i}}{2\delta\omega} = 0.6 + \frac{35}{2 \cdot 0.015 \cdot 314.2} = 0.6 + 3.71 = 4.31 \quad (12.7\,\mathrm{dB}) \tag{6.3}$$

and a -3 dB width of the peak $2\delta\omega = 9.4\,\mathrm{rad/s}$ (1.5 Hz), so the controller still has a high gain when the grid frequency moves by a fraction of a hertz. The resonator alone is a second-order system with poles at $s = -\delta\omega \pm j\omega\sqrt{1-\delta^2} = -4.7 \pm j314.1$ rad/s: its free response decays by $e^{-2\pi\delta} = 0.91$ every grid period. The block uses $\delta = 0.005$ (Section 6.3): peak $0.6 + 11.14 = 11.74$ (21.4 dB), width 0.5 Hz, decay 0.97 per period.

The code writes the resonator in the state-space form that the model's class `ctrl_afe_setup` uses (fields `Ares`, `Bres`, `Cres` of `res_pi`):

$$\dot{x}_1 = x_2, \qquad \dot{x}_2 = -\omega^2 x_1 - 2\delta\omega\, x_2 + k_\mathrm{i}\, e, \qquad y = x_2, \qquad d = k_\mathrm{p} e + x_2, \tag{6.4}$$

that is $A = \begin{bmatrix} 0 & 1 \\ -\omega^2 & -2\delta\omega \end{bmatrix}$, $B = \begin{bmatrix} 0 \\ k_\mathrm{i} \end{bmatrix}$, $C = [0\ 1]$. The transfer function from e to x_2 is $k_\mathrm{i} s/(s^2 + 2\delta\omega s + \omega^2)$, the resonant term of (6.2). (The class keeps k_i outside, $B = [0; 1]$ with a gain in the diagram; the code puts it in the update, which is the same thing.)

6.3 The discrete implementation, as in the code

The block updates the two states of each phase once per control period with a forward Euler step on x_1 and a backward Euler step on x_2: the *updated* x_1 is used in the update of x_2 (library rev. 03; rev. 00 to 02 used the plain forward Euler of the model, Section 6.4). The lines of `current_ctrl_uvw.m` are, for the three phases at once (`x_res` is 2×3, `e` is 3×1):

```
e = i_abc_ref - is_abc;                                  % [3x1]
d_abc = kp*e + x_res(2, :)';                             % output: P + resonator state x2
x1 = x_res(1, :) + ts*x_res(2, :);                       % new x1 ...
x2 = x_res(2, :) + ts*(-w^2*x1 - 2*delta*w*x_res(2, :) + ki*e');   % ... used in the update of x2
x_res(1, :) = x1;
x_res(2, :) = x2;
```

so the output of the step uses the state x_2 *before* the update (no direct feed-through of e into the resonator part, the controller is strictly proper apart from k_p), and the states move afterwards:

$$x_1^+ = x_1 + t_\mathrm{s}\, x_2, \qquad x_2^+ = x_2 + t_\mathrm{s}\left(-\omega^2 x_1^+ - 2\delta\omega\, x_2 + k_\mathrm{i}\, e\right). \tag{6.5}$$

In the z domain the controller is

$$C_\mathrm{d}(z) = k_\mathrm{p} + \frac{k_\mathrm{i}\, t_\mathrm{s}\,(z-1)}{z^2 - (2 - \omega^2 t_\mathrm{s}^2 - 2\delta\omega t_\mathrm{s})\, z + 1 - 2\delta\omega t_\mathrm{s}}, \tag{6.6}$$

whose poles are the roots of the denominator. The product of the two complex roots is the constant term, so the radius of the poles is

$$|z_\mathrm{res}| = \sqrt{1 - 2\delta\omega t_\mathrm{s}} \approx 1 - \delta\omega t_\mathrm{s}, \tag{6.7}$$

the radius $e^{-\delta\omega t_\mathrm{s}}$ of the exact mapping to first order: exactly on the unit circle for $\delta = 0$ (a lossless resonator, which can now be used safely) and inside it for any $\delta > 0$. The resonance



is at 50.007 Hz with $\delta = 0.005$ (the forward step does not warp the frequency noticeably at $\omega t_s = 0.039$, so no pre-warping is needed). This is the discretization that [6] recommends for the two-integrator form; Tustin with pre-warping at ω gives the same poles with more arithmetic.

The damping of the block is $\delta = 0.005$ (`delta_res` of the example structure and of the model init), not the 0.015 of the model's class, for a reason that Section 6.4 explains: the gains of the model were tuned with a forward Euler resonator whose gain at the grid frequency is 10.8, about three times the 3.7 of the continuous design with $\delta = 0.015$, and $\delta = 0.005$ keeps that gain with the new step ($k_i/(2\delta\omega) = 11.1$; the exact match would be $\delta = 0.0052$). The numbers of the block, and those of the same step with the damping of the model:

forward-backward Euler resonator, $t_s = 125\,\mu s$	$\delta = 0.005$ (the block)	$\delta = 0.015$
poles	$0.999033 \pm j\,0.039258$	$0.998640 \pm j\,0.039246$
radius $\lvert z_{res}\rvert$ by (6.7)	0.999804	0.999411
resonance frequency (angle of the poles divided by $2\pi t_s$)	50.007 Hz	50.012 Hz
decay of the free response per grid period, $\lvert z_{res}\rvert^{160}$	0.969	0.910
gain $\lvert C_d(e^{j\omega t_s})\rvert$ at 50 Hz (resonator alone)	11.74 (11.14)	4.31 (3.71)
−3 dB width of the peak, $2\delta\omega$	0.5 Hz	1.5 Hz

With $\delta = 0.005$ the resonator is a lightly damped oscillator: left alone (zero error), its 50 Hz oscillation decays by 3 % every period (time constant $1/(\delta\omega) = 0.64$ s, Figure 5, right), its gain at the grid frequency is 11.7 (21.4 dB, Figure 4) and its peak is 0.5 Hz wide at −3 dB, which is the width the model's resonator had too (its poles are as close to the circle, $1.8 \cdot 10^{-4}$ outside against $2.0 \cdot 10^{-4}$ inside). The resonator is tuned at the nominal pulsation (`omega_res`), not at the one of the PLL: with a grid frequency 0.25 Hz off the gain of the resonator drops to 0.71 of its peak, still 7.9, and the current loop keeps its 50 Hz notch within a few dB (Section 6.5).

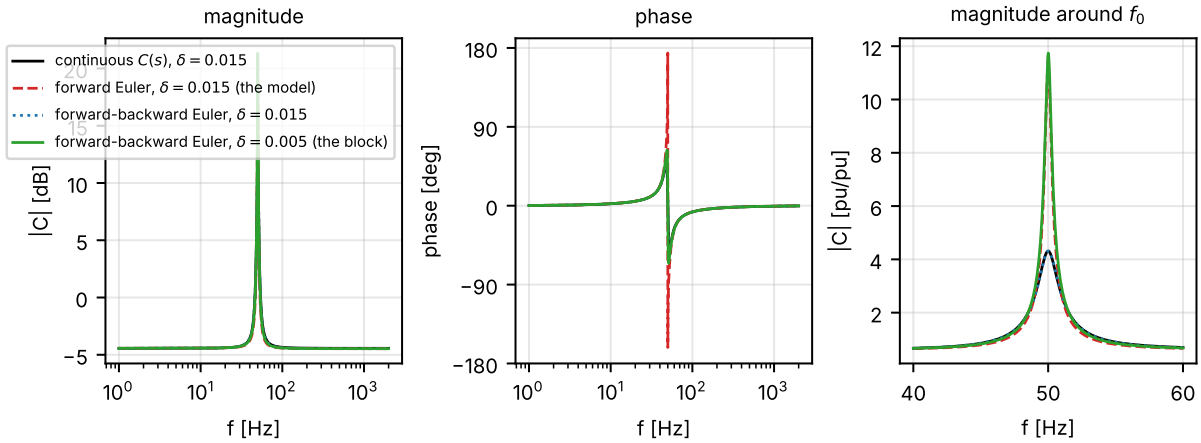


Figure 4: Frequency response of the resonant PI, $k_p = 0.6$, $k_i = 35/s$: continuous $C(s)$ with $\delta = 0.015$, the forward Euler resonator of the model ($\delta = 0.015$), the forward-backward Euler resonator of the block with $\delta = 0.015$ and with its $\delta = 0.005$. Right: the peak at f_0, 4.3 for the continuous design and for the forward-backward step with $\delta = 0.015$, 11.4 for the model and 11.7 for the block. Script `afe_uvw_ctrl_figures.py`.



6.4 The forward Euler resonator of the model

The class `ctrl_afe_setup` of the model discretizes the same state space with the plain forward Euler step (`Aresd = eye(2) + Ares*ts`, `Bresd = Bres*ts` in `resonant_pi`), that is, with the *old* x_1 in the update of x_2:

$$x_1^+ = x_1 + t_\mathrm{s}\, x_2, \qquad x_2^+ = x_2 + t_\mathrm{s}\left(-\omega^2 x_1 - 2\delta\omega\, x_2 + k_\mathrm{i}\, e\right), \tag{6.8}$$

and the block did the same up to library rev. 02. The characteristic polynomial is then $z^2 - (2 - 2\delta\omega t_\mathrm{s})\, z + 1 - 2\delta\omega t_\mathrm{s} + \omega^2 t_\mathrm{s}^2$ and the radius of the poles

$$|z_{\mathrm{res,FE}}| = \sqrt{1 - 2\delta\omega t_\mathrm{s} + \omega^2 t_\mathrm{s}^2}. \tag{6.9}$$

The forward Euler step adds the term $\omega^2 t_\mathrm{s}^2$, which pushes the poles outwards: this resonator is stable only when

$$\delta > \frac{\omega t_\mathrm{s}}{2} = \frac{314.16 \cdot 125 \times 10^{-6}}{2} = 0.0196. \tag{6.10}$$

With the damping of the model, $\delta = 0.015$, the condition is not met:

poles of the forward Euler resonator, $\delta = 0.015$, $t_\mathrm{s} = 125\,\mu\mathrm{s}$	$0.999411 \pm j\,0.039265$
radius $\lvert z_{\mathrm{res,FE}}\rvert$ by (6.9)	1.000182
resonance frequency (angle of the poles divided by $2\pi t_\mathrm{s}$)	49.998 Hz
equivalent continuous damping, $-\ln\lvert z_{\mathrm{res,FE}}\rvert/(\omega t_\mathrm{s})$	−0.0046
growth of the free response per grid period, $\lvert z_{\mathrm{res,FE}}\rvert^{160}$	1.030
gain $\lvert C_\mathrm{d}(e^{j\omega t_\mathrm{s}})\rvert$ at 50 Hz (resonator alone)	11.42 (10.82)

The resonator of the model is a slightly *growing* oscillator: left alone (zero error), its 50 Hz oscillation grows by 3 % every period instead of decaying by 9 % (Figure 5, right). The resonance frequency is practically exact, but the peak gain is 11.4 instead of the 4.3 of the continuous design, because the poles are closer to the unit circle than the continuous poles are to the $j\omega$ axis (Figure 4); the phase of C_d wraps through 180° at the resonance, which is the signature of a pole outside the unit circle. The gains of the model (k_p, k_i, δ) were tuned with this resonator, so the gain 10.8 at the grid frequency is what the current loop of the model has, and it is the gain the block keeps: the forward-backward step with $\delta = 0.015$ would cut the gain at 50 Hz from 11.4 to 4.3 and the depth of the notch of the loop from 41 dB to 33 dB (Section 6.5), while $\delta = 0.005$ gives 11.7 and 41 dB with the poles inside the circle.

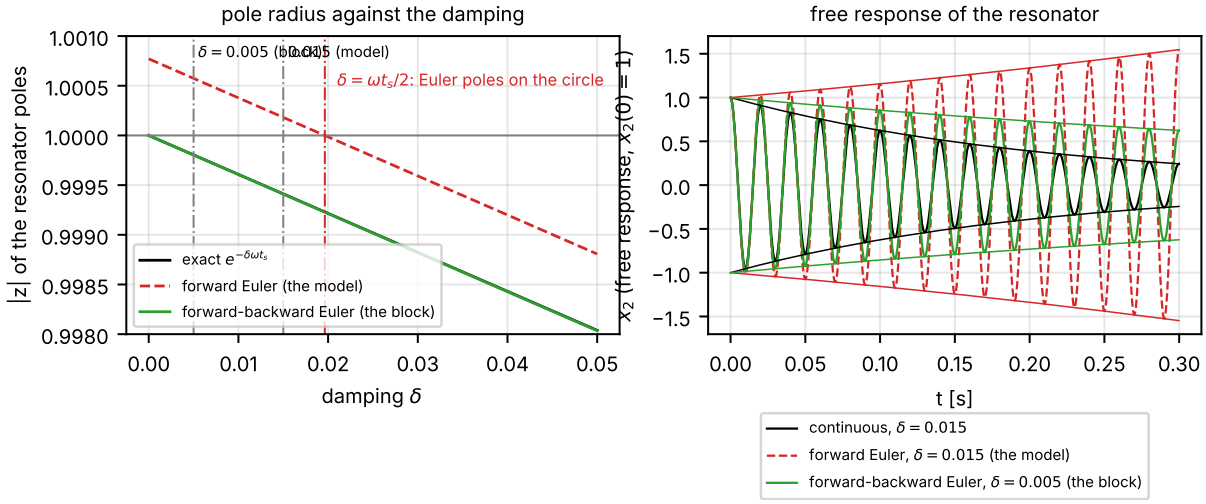


Figure 5: Left: radius of the resonator poles against the damping δ for the exact mapping, the forward Euler of the model and the forward-backward Euler of the block (which coincides with the exact curve at this scale); the forward Euler poles are outside the unit circle for $\delta < \omega t_s/2 = 0.0196$. Right: free response of the resonator state x_2 from $x_2(0) = 1$: the continuous resonator with $\delta = 0.015$ decays by 9 % per period, the forward Euler one of the model grows by 3 %, the block ($\delta = 0.005$) decays by 3 %.

6.5 What it means for the current loop

A growing resonator does not make the loop unstable by itself: the poles of the closed loop are what counts, and the loop gain at 50 Hz moves them inside the circle. To show it with numbers, the loop is closed on the plant of the test bench of the package: three RL branches between the converter and the grid in pu, $x_\mathrm{l} = 0.1$ (reactance at f_0) and $r = 0.01$, integrated exactly over one step,

$$P(z) = \frac{K}{z - e_\mathrm{p}}, \qquad e_\mathrm{p} = e^{-r\omega t_\mathrm{s}/x_\mathrm{l}}, \quad K = \frac{1 - e_\mathrm{p}}{r}, \qquad L(z) = C_\mathrm{d}(z)\, z^{-1}\, P(z), \tag{6.11}$$

where z^{-1} is the one-step output delay of the block (Section 6.6). The plant of the model (690 V, 250 kW, 500 µH: $x_\mathrm{l} = 0.092$ pu, $r = 0.0023$ pu) gives almost the same numbers. The results of the companion script for the block, for the resonator of the model and for the forward-backward step with the damping of the model (FE = forward Euler, FB = forward-backward Euler):

plant	resonator	crossover	phase margin	largest closed-loop $\lvert z\rvert$	$\lvert S\rvert$ at 50 Hz
bench	the block (FB, $\delta = 0.005$)	300 Hz	68.9°	0.99617	0.0086 (−41.3 dB)
bench	the model (FE, $\delta = 0.015$)	300 Hz	68.9°	0.99656	0.0088 (−41.1 dB)
bench	FB, $\delta = 0.015$	300 Hz	68.9°	0.99576	0.0233 (−32.7 dB)
model 690 V	the block (FB, $\delta = 0.005$)	327 Hz	66.5°	0.99612	0.0078 (−42.1 dB)
model 690 V	the model (FE, $\delta = 0.015$)	327 Hz	66.5°	0.99651	0.0080 (−41.9 dB)
model 690 V	FB, $\delta = 0.015$	327 Hz	66.5°	0.99571	0.0213 (−33.4 dB)

The closed loop is stable with a good margin in every case; the crossover (about 300 Hz, set by $k_\mathrm{p}\omega_\mathrm{nom}/x_\mathrm{l}$) and the margin do not depend on the resonator, which is small away from 50 Hz. With the block, the resonant poles of the closed loop sit at $\lvert z\rvert = 0.9962$ (a time constant of

33 ms) and the sensitivity at 50 Hz is 0.9 %: a 50 Hz disturbance (the grid voltage, when it is not fed forward by the modulator) is attenuated by 41 dB, and the steady-state tracking error of a 50 Hz reference is below 1 % of its amplitude. This is the same loop as with the resonator of the model (0.9 %, 36 ms) and three times better than what the forward-backward step with $\delta = 0.015$ would give (2.3 %). The test bench confirms the order of magnitude: the stack current reaches its -0.85 pu reference within 6×10^{-3} pu with either resonator (Section 9).

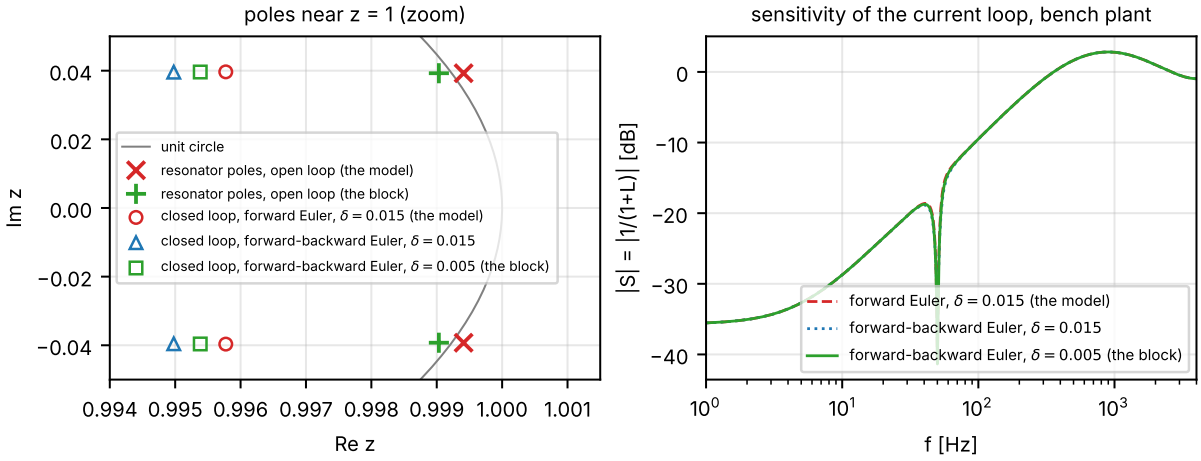


Figure 6: Per-phase current loop on the bench plant ($x_1 = 0.1$, $r = 0.01$, one step of delay). Left: the open-loop resonator poles of the model (outside the circle, radius 1.000182) and of the block (inside, 0.999804), and the closed-loop poles with the three resonators: the loop pulls the resonant pair inside the circle in every case. Right: sensitivity $|S| = |1/(1 + L)|$; the notch at 50 Hz is 41 dB deep with the block and with the resonator of the model, 33 dB with the forward-backward step at $\delta = 0.015$.

Two consequences are worth keeping in mind:

- the damping of the resonant mode of the loop depends on the loop gain at 50 Hz. When the vector limiter of Section 6.6 divides the output by `scale`, the loop gain drops by the same factor: with the bench plant the largest closed-loop radius grows from 0.99617 (scale 1) to 0.99765 (scale 5) and 0.99888 (scale 10, the largest allowed), still inside the circle, but with a mode that decays about three times more slowly. With the resonator of the model it was 0.99927 at scale 10, closer to the circle, because its open-loop poles start outside it;
- the resonator states have no anti-windup: while the limiter is active, or while the modulator is disabled, x_1 and x_2 keep integrating the error with their own dynamics. With the block these dynamics decay (3 % per period, 0.64 s); with the resonator of the model they grew by 3 % per period. In both cases a long time with the gates off and a non-zero current error should be avoided, or the states should be reset (the block has no reset input: the subsystem can be re-initialized, or a reset can be added like the one of `frt_vde4110`).



6.6 Vector limiter, transformation to $\alpha\beta$ and output delay

The three outputs d_{abc} of the resonant PIs are divided by one common factor,

$$\mathrm{scale} = \mathrm{sat}\big(|u_{\alpha\beta}(k-1)|,\ \texttt{mag_lim},\ \texttt{mag_scale_max}\big), \qquad d_{\mathrm{abc}} \leftarrow d_{\mathrm{abc}}/\mathrm{scale}, \tag{6.12}$$

where $|u_{\alpha\beta}(k-1)|$ is the length of the $\alpha\beta$ output vector of the previous step. With `mag_lim` $= 1$ and `mag_scale_max` $= 10$ the factor is 1 as long as the voltage vector is inside the unit circle (the linear range of the modulator) and grows up to 10 when the controller asks for more: the vector is scaled, not clipped per phase, so its direction is kept. This is the `antiwind_up` subsystem of the model. It acts one step late (it reads the previous output) and, as said above, it does not touch the resonator states.

The scaled outputs are transformed to $\alpha\beta$ with (2.1), which drops their zero sequence (the three resonant PIs are independent and their outputs need not sum to zero; the zero sequence of the voltage is set by the next loop instead). The result is delivered one step late when `out_delay` is 1, as the Unit Delay blocks of the model do: the previous $u_{\alpha\beta}$ is output and the new one stored. The delay is part of the loop analysis of Section 6.5; `out_delay` $= 0$ removes it, for a model whose modulator already accounts for the computation delay.

6.7 Zero-sequence current loop

The third component of the output, u_γ, is set by a PI with clamped integrator (5.2) on the zero-sequence current,

$$i_{\mathrm{z}} = \tfrac{1}{3}\left(i_{\mathrm{s,a}} + i_{\mathrm{s,b}} + i_{\mathrm{s,c}}\right), \qquad e_{\mathrm{z}} = \texttt{iz_ref} - i_{\mathrm{z}}, \qquad u_\gamma = \mathrm{sat}\big(k_{\mathrm{p,z}} e_{\mathrm{z}} + u_{\mathrm{i,z}},\ \pm\texttt{u_z_lim}\big), \tag{6.13}$$

with $k_{\mathrm{p,z}} = 0.5$, $k_{\mathrm{i,z}} = 18/\mathrm{s}$, the limit 1 pu and the reference 0. It cancels a common-mode current of the stack, for instance the one driven by a DC offset of the grid branches or by an asymmetry of the modulation. In the test bench a 0.02 pu DC offset in the branch voltages is cancelled to 7×10^{-13} pu.

6.8 Log output

`is_xe` $= [\xi;\ \eta]$ is the measured stack current in the positive frame (Clarke and rotation by $+\theta$), the `id_xi_pu`, `id_eta_pu` signals of the model, for the scopes. It is not used by the control.

7 Islanding detector (islanding_detector)

This block is new: the model had a subsystem that only measured the 7th harmonic of the grid voltage. Inputs: `ug_ab` (from Section 3), `theta`, `omega_pu`, `enable`. The detector is *passive*: it watches what the voltage at the point of connection does when the grid is lost and the converter keeps feeding the local load, and it trips when one of six criteria holds for its whole on-delay (Table 2).



Measurements. The fundamental and two harmonics are measured by turning $\alpha\beta$ with a multiple of the PLL angle and averaging over one grid period (the same moving average (2.4) with $k_{\text{avg}} = $ `h_avg_order` $= 1$, which removes every other harmonic):

$$\begin{aligned} &\text{fundamental, positive frame:} && \alpha c_1 + \beta s_1, && -\alpha s_1 + \beta c_1, \\ &\text{5th harmonic, negative frame:} && \alpha c_5 - \beta s_5, && \alpha s_5 + \beta c_5, \qquad c_h = \cos h\theta,\ s_h = \sin h\theta, \\ &\text{7th harmonic, positive frame:} && \alpha c_7 + \beta s_7, && -\alpha s_7 + \beta c_7, \end{aligned} \tag{7.1}$$

and u_1, u_5, u_7 are the lengths of the three averaged pairs. The harmonic criterion is the ratio $h = \sqrt{u_5^2 + u_7^2}\,/\max(u_1, u_{\text{valid}})$, with $u_{\text{valid}} = $ `u_valid`: the 5th and the 7th are the harmonics that an inverter with dead time or a non-linear local load produces most, and they rise when the stiff grid is gone. The frequency is the PLL pulsation through a first-order low pass with time constant `tau_f` (20 ms), $f_{\text{lpf}} \leftarrow f_{\text{lpf}} + \frac{t_s}{\tau_f + t_s}(\omega_{\text{pu}} - f_{\text{lpf}})$, in pu; the ROCOF is the rate state of a double-integrator observer (4.1) on ω_{pu} with poles at $[-20,\ -80]$ rad/s ($k_x = 99.6$/s, $k_v = 1590/\text{s}^2$), scaled to Hz/s.

Table 2: Criteria of the islanding detector with the default thresholds (the network protection values of VDE-AR-N 4105 / 4110 [4, 3], plus ROCOF and harmonics). Each criterion has its bit in `crit`.

bit	criterion	condition	on-delay	needs $u_1 \geq$ `u_valid`
1	under-voltage	$u_1 <$ `u_min` $= 0.8$ pu	`t_trip_u_min` $= 2.7$ s	no
2	over-voltage	$u_1 >$ `u_max` $= 1.15$ pu	`t_trip_u_max` $= 0.1$ s	no
4	under-frequency	$f_{\text{lpf}} <$ `f_min` $= 47.5$ Hz	`t_trip_f` $= 0.1$ s	yes
8	over-frequency	$f_{\text{lpf}} >$ `f_max` $= 51.5$ Hz	`t_trip_f` $= 0.1$ s	yes
16	ROCOF	$\lvert \mathrm{d}f/\mathrm{d}t \rvert >$ `rocof_max` $= 2.5$ Hz/s	`t_trip_rocof` $= 0.1$ s	yes
32	harmonics	$h >$ `h_max` $= 0.1$	`t_trip_h` $= 0.1$ s	yes

Logic. A timer per criterion counts the time during which its condition holds and is reset when it does not; the bit is set when the timer reaches the on-delay. The frequency, ROCOF and harmonic criteria are blocked while $u_1 <$ `u_valid` (0.5 pu): a PLL in a deep voltage dip is not a frequency meter, and a grid fault is the business of the FRT block, not of this one. The detector is armed when `enable` is high and `t_enable` (0.1 s) has passed; with `latch` $= 1$ the output `island` stays 1 from the first trip until `enable` goes to 0, otherwise it follows the criteria. The log gives u_1, u_5, u_7, h, f_{lpf}, the ROCOF, the longest running timer and the time since the detection.

Limits. When the local load takes exactly the active and the reactive power of the converter, voltage and frequency do not move after the loss of the grid and only the harmonics can tell (the non-detection zone of every passive method). The slow under-voltage stage (0.8 pu for 2.7 s) is what lets the converter ride through a grid fault without an islanding trip: in the test bench the dip of 150 ms does not trip the detector, while a frequency ramp of 6 Hz/s (the grid lost with a surplus of active power) trips the ROCOF criterion 0.14 s after its start (Figure 8). In the model the block is in the library but not yet wired (Section 11).



8 Parameter structure and the model init

All five blocks read the same structure `p`: the common fields at the top level and one substructure per block (Table 3). The file `afe_uvw_ctrl_param_example.m` of the package builds it with the numbers of the model, `p = afe_uvw_ctrl_param_example(ts, f_nom)`, and computes the observer gains with (4.2). In the model the section "AFE control in uvw" of `init_model.m` builds `param.afe.uvw` field by field from the existing objects of the init (`param.afe.ctrl` from the class `ctrl_afe_setup`, `param.frt.data`, `param.ctrl`, `param.glb_time`, `param.grid`), so that a change of a gain in the init reaches the blocks; the section comes after "Control settings common to all modules" because it reads `param.ctrl`. The last column of the table says where each number comes from in the model.

Table 3: The parameter structure `p` (`param.afe.uvw` in the model): fields, meaning, values of the model and their source in `init_model.m` (`c` = `param.afe.ctrl`).

field	meaning	value	source in the model
common			
`ts`	control period [s], integration step of all blocks	125×10^{-6}	`param.glb_time.ts_afe`
`omega_nom`	nominal grid pulsation [rad/s], base of `omega_pu`	$2\pi\,50$	`param.grid.emu.omega_grid_nom`
vseq (grid voltage sequences) and **iseq** (grid current sequences), same four fields			
`avg_order`	window = grid period / `avg_order` (2: half period)	2	`param.ctrl.mavarage_filter_frequency_base_order`
`omega_pu_min`, `_max`	limits of the measured pulsation used for the window [pu]	0.1, 2	constants of the model's saturation
`t_enable`	averages are zero before this time [s]	0.025	`param.ctrl.dmavg_filter_enable_time`
iseq, the three PIs			
`kp_pos`, `ki_pos`	PI of the positive-sequence η grid current, [–], [1/s]	0.25, 35	`c.kp_rc_pos_grid`, `c.ki_rc_pos_grid`
`kp_neg`, `ki_neg`	PI of the negative-sequence ξ and η grid currents	0.25, 35	`c.kp_rc_neg_grid`, `c.ki_rc_neg_grid`
`lim_pos_eta`	limit of the stack current reference pos_eta [pu]	1.0	`c.ieta_pos_ref_lim`
`lim_neg_xi`	limit of neg_xi [pu] (the model uses the positive limit)	1.0	`c.ieta_pos_ref_lim`
`lim_neg_eta`	limit of neg_eta [pu]	0.5	`c.ieta_neg_ref_lim`
`err_sign`	-1: grid current measured towards the converter	-1	the gain -1 of the model's diagram
frt			
`obs_kx`, `obs_kv`	gains of the voltage-rate observers [1/s], [1/s^2]	156.0, 3909	`c.vr_obs.kx`, `c.vr_obs.kv` (poles $[-0.1, -0.4]\,\omega$)
`t_obs_release`	observers held at zero before this time [s]	0.025	`time_release` of the model
`v_pos_low`, `v_pos_high`	band of the positive sequence [pu]	0.85, 1.15	`LVRT_VPOS_TH_LOW`, `LVRT_VPOS_TH_HIGH`
`v_hyst`	band narrowing to leave the fault [pu]	0.1	`LVRT_SEQ_DETECTION_HYST`



field	meaning	value	source in the model
v_neg_max	negative sequence above this is a fault [pu]	0.1	LVRT_VNEG_TH
v_pos_dot_max, v_neg_dot_max	voltage rate needed for a detection [pu/s]	2, 2	LVRT_VPOS_DOT_TH, LVRT_VNEG_DOT_TH
t_rearm	idle time before a new detection [s]	0.125	1000 samples of the model
t_hold_max	fault longer than this: EXCEEDED [s]	7.625	61000 samples of the model
pwrlim_v_pos, pwrlim_v_neg	power limitation when pos_xi below / $\lvert v_{\mathrm{neg}}\rvert$ above [pu]	0.8, 0.1	LVRT_VPOS_PWRLIM_TH, LVRT_VNEG_PWRLIM_TH
pwrlim_hyst	release when pos_xi > pwrlim_v_pos + this [pu]	0.1	the model's diagram
pwrlim_v_neg_release	... and $\lvert v_{\mathrm{neg}}\rvert$ below this [pu]	0.7	the model's diagram (see open points)
reddyn_enabled	reduced dynamic support on/off	0	constant 0 of the model
reddyn_v_pos, reddyn_v_neg	thresholds of the reduced support [pu]	0.7, 0.2	LVRT_VPOS_RED_DYN_TH, LVRT_VNEG_RED_DYN_TH
k_frt_ref	reactive current per voltage change [pu/pu]	2	param.frt.data.k_frt_ref
t_avg_prefault	window of the pre-fault averages [s]	0.5	avg_flt_500ms of the model
t_avg_start	the pre-fault averages start here [s]	0.05	new (the model started at 0)
lim_pos_eta, lim_neg_xi, lim_neg_eta	limits of the grid current references [pu]	1.0, 0.5, 0.5	c.i_grid_pos_eta_lim, c.i_grid_neg_xi_lim, c.i_grid_neg_eta_lim
lim_*_red	the same with reduced dynamic support [pu]	0.1	c.i_grid_*_red_lim
f_lpf_ref	low pass of the references [Hz]	161	g0_161Hz of the model
ictrl			
kp, ki	resonant PI: proportional gain [–] and resonant gain [1/s]	0.6, 35	c.res_pi.kp_rpi, c.res_pi.ki_rpi (LVRT setting of the init; class defaults 0.2, 45)
delta_res	damping of the resonator (forward-backward Euler, Section 6.3)	0.005	0.005 in the init; the class has 0.015 for its forward Euler resonator (Section 6.4)
omega_res	resonance pulsation [rad/s]	$2\pi\,50$	c.omega_base
mag_lim, mag_scale_max	vector limiter: length above which the output is scaled, largest factor	1, 10	constants of antiwind_up
out_delay	1: $\alpha\beta$ output one step late (Unit Delays of the model)	1	–
kp_z, ki_z	zero-sequence current PI [–], [1/s]	0.5, 18	c.kp_is_z, c.ki_is_z
u_z_lim, iz_ref	limit of u_γ [pu], zero-sequence reference [pu]	1, 0	constants of the model
island (new block: all values are defaults of the library)			
h_avg_order	harmonic averaging window = grid period / this	1	
omega_pu_min, _max	limits of the measured pulsation for the window [pu]	0.5, 1.5	
t_enable	no detection before this time [s]	0.1	
tau_f	low pass of the frequency [s]	0.02	
obs_poles, obs_kx, obs_kv	ROCOF observer: poles [rad/s] and gains [1/s], [1/s^2]	$[-20, -80]$, 99.6, 1590	the init computes the gains with acker



field	meaning	value	source in the model
u_valid	frequency and harmonic criteria only above this voltage [pu]	0.5	
u_min, t_trip_u_min	under-voltage [pu], [s]	0.8, 2.7	VDE-AR-N 4105 / 4110
u_max, t_trip_u_max	over-voltage [pu], [s]	1.15, 0.1	VDE-AR-N 4105 / 4110
f_min, f_max, t_trip_f	frequency band [pu], delay [s]	47.5/50, 51.5/50, 0.1	VDE-AR-N 4105 / 4110
rocof_max, t_trip_rocof	ROCOF [Hz/s], delay [s]	2.5, 0.1	
h_max, t_trip_h	harmonic ratio [–], delay [s]	0.1, 0.1	
latch	1: output stays 1 until enable goes to 0	1	

9 How the code was checked

Python twin of the model's diagrams. The script tools/check_afe_uvw_ctrl.py of the package holds a twin, in Python and without MATLAB, of the Simulink diagrams that were replaced (dmavgflt_process, LVRT_detector, linear_double_integrator_observer, avg_flt, u_161Hz_flt, the resonant PIs, antiwind_up, zero_sequence_current_ctrl, written block by block from the .slx and from the library sources) and runs it together with the five functions (in GNU Octave) on the same inputs for 1.2 s: a 50.2 Hz grid with 2 % 7th and 1.5 % 5th harmonic, a dip to 0.55 pu with 0.25 pu negative sequence at 0.75–0.90 s, steps of the grid current and of the references. The comparison is made block by block (the twin reads the signals of the new code at the block borders, so that each block sees the same inputs): the transformations agree to 4×10^{-16}, the observers to 7×10^{-14}, the PIs of the reactive loops to 4×10^{-14}, the current controller (u_abg, i_abc_ref) to 2×10^{-13} (the twin uses the forward-backward resonator of the block; its option --model-resonator switches it to the forward Euler of the model with $\delta = 0.015$, for a comparison with the model as it is); the sequence averages differ by 2×10^{-4} in steady state and 3×10^{-3} at the edges of the dip (a different moving average, Section 10), the FRT references by 5×10^{-4} (pre-fault average one sample apart), and the two implementations detect the fault at the same samples (0.754 88 s, 0.908 13 s). Figure 7 shows the chain run.

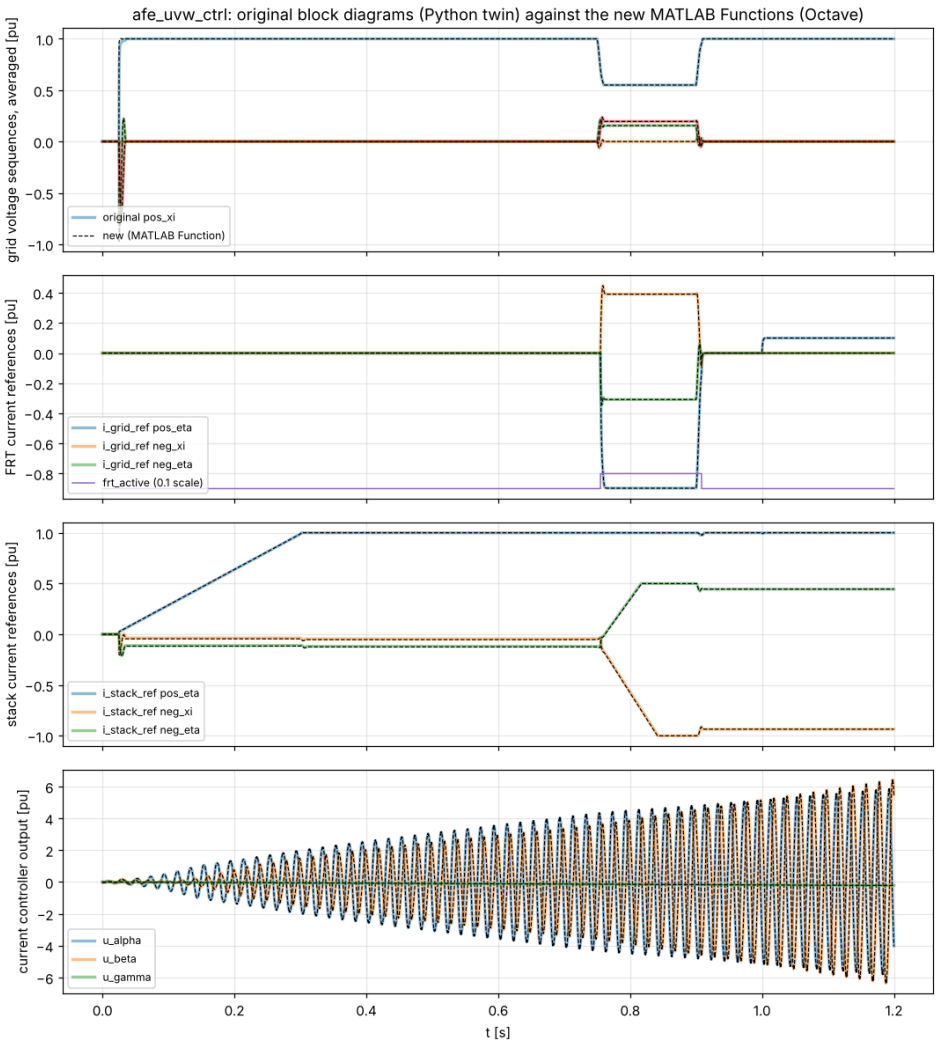


Figure 7: The Python twin of the original block diagrams (solid) against the five MATLAB Functions run in GNU Octave (dashed), each implementation on its own signals, open loop: averaged voltage sequences, FRT references, stack current references of the reactive loops and the output of the current controller (which grows because the loop is open: the resonators integrate the error of a current that never comes).

The test bench in a loop. `test/afe_uvw_ctrl_test_loop.m` runs the five functions in a loop with a source (grid voltage with harmonics, PLL angle, references) and the RL plant of (6.11) behind a unit delay, 1.6 s; `afe_uvw_ctrl_test_run.m` runs the same scenario in the Simulink bench `test_afe_uvw_ctrl.slx` and in the loop and prints 15 checks. In GNU Octave the 14 behaviour checks pass, with the forward-backward resonator and $\delta = 0.005$ as with the forward Euler one before: fault detected 5 ms after the start of the dip and released 8 ms after its end, power limitation set, FRT reference -0.80 pu reached within 3×10^{-3}, grid η current on its reference within 5×10^{-5} before the dip and within 0.02 at the end of the dip, stack ξ current on -0.85 pu within 6×10^{-3}, zero-sequence current 7×10^{-13}, islanding detected 0.14 s after the start of the 6 Hz/s ramp by the ROCOF criterion and not by the dip (Figure 8). In MATLAB R2026b all 15 checks passed with rev. 02 of the library, the 15th being "blocks = functions":

the Simulink bench and the loop give the same signals (difference 0); the run with the resonator of rev. 03 is pending.

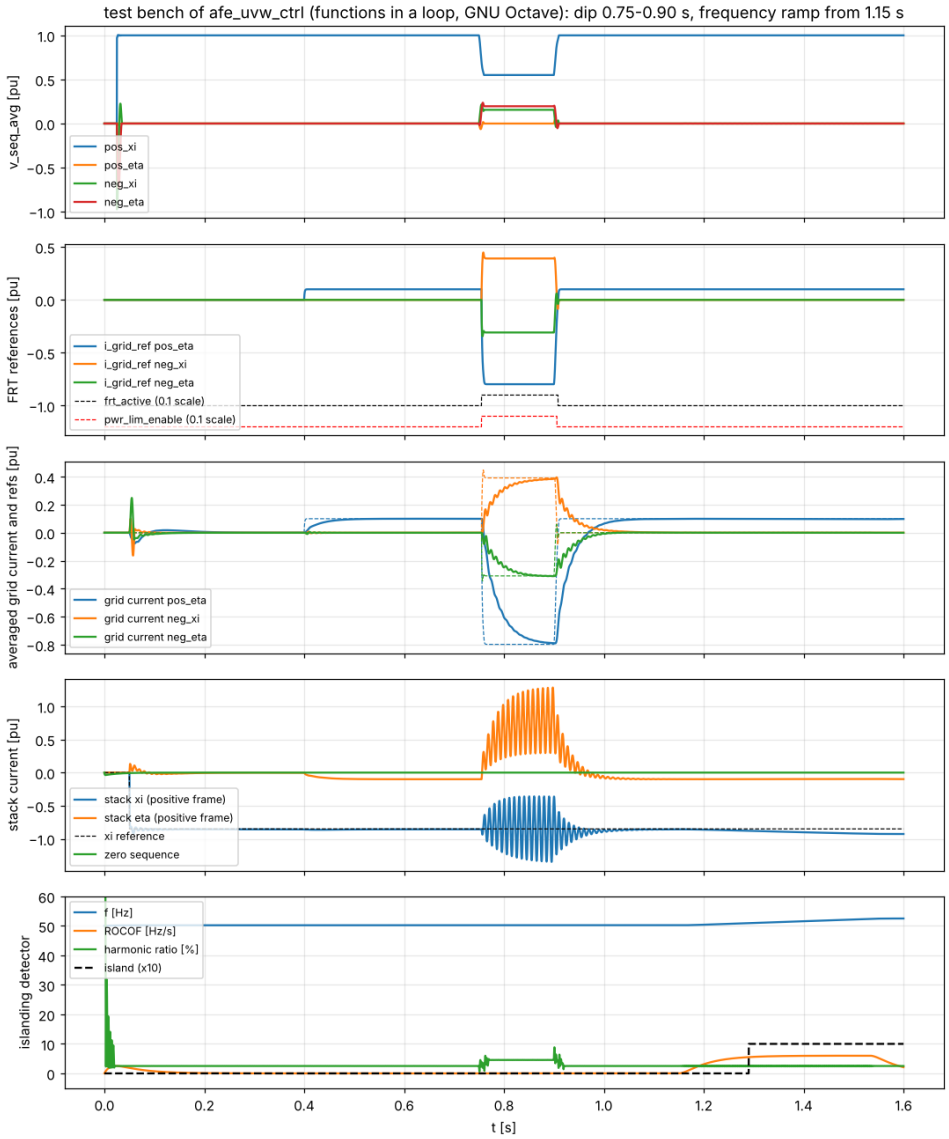


Figure 8: Test bench of the package run as a loop in GNU Octave: averaged voltage sequences, FRT references and flags, averaged grid current against its references, stack current in the positive frame with the zero sequence, and the islanding detector (frequency, ROCOF, harmonic ratio, trip). Dip at 0.75–0.90 s, frequency ramp of 6 Hz/s from 1.15 s.

In the model. `afe_rst_ctrl.slx` of `afe_uvw_three_phase` now uses the five blocks inside `afe_control`, all with the mask parameter `param.afe.uvw` (Section 8); the model compiles with the explicit port sizes of the blocks (library PR #35). The runs of the model with the new blocks belong to the model, not to this note.



10 Differences from the Simulink subsystems they replace

- **Resonator discretization.** The block uses the updated x_1 in the update of x_2 (forward-backward Euler, Section 6.3) since library rev. 03; the class of the model uses forward Euler, whose poles are outside the unit circle for $\delta < \omega t_s/2$ (Section 6.4). The damping is 0.005 instead of 0.015, so that the gain at the grid frequency and the current loop stay the same.
- **Moving average.** `dmavgflt_process` (the MATLAB port of the C `mavgflt`) averages the samples up to the previous one and starts from an implicit zero sample; `moving_average_frac` includes the present sample and is unbiased while the window fills up. Same steady-state value (within the 2ω residual, 2×10^{-4}), one sample of lead, no start-up transient.
- **Pre-fault averages of the FRT** start at `t_avg_start` (50 ms) instead of 0: the model's average takes the zeros and the start-up transient of the sequence averages into its 0.5 s window, which puts an offset of about 1×10^{-2} on the negative-sequence references of a fault in the first second.
- **`frt_state` in the log** is the state after the update of the step (the model reported the state of the previous step); `frt_active` and `pwr_lim_enable` have the same timing as the model.
- **Power-limitation release** on $|v_{\mathrm{neg}}| < 0.7$, as in the model's diagram, where the threshold used is `LVRT_VPOS_PWRLIM_TH` minus the hysteresis (probably a slip); the value is the parameter `pwrlim_v_neg_release`.
- **Reduced dynamic support** is a parameter (default off) instead of the constant 0 of the model.
- **Islanding detector** is new; the model only measured the 7th harmonic.
- **Port sizes** of the MATLAB Functions are explicit (the model could not resolve the inherited sizes of `i_abc_ref`).
- The feed-forward of the grid voltage, the DC-link control, the PLL and the modulator are not part of these blocks, as they were not part of the replaced subsystems.

11 Open points

1. **Direction of the reactive current support.** With the η axis of (2.2) (the usual q axis) and the grid voltage on $+\xi$, a current measured *into* the converter that leads the voltage by 90° (capacitive, the support that VDE-AR-N 4110 asks for in an under-voltage) has a *positive* η component. The FRT rule (4.3) gives a *negative* η reference in a dip, so the support is capacitive only if the grid current of the model is measured *out of* the converter, towards the grid; with the grid current measured towards the converter (as the sign of the reactive loops suggests, Section 5) it would be inductive. The code reproduces the model exactly here (the twin agrees to 4×10^{-13}), so this is a check to make once on the model with the real plant: during a dip with `i_react_ref` $= 0$, the positive-sequence voltage at the connection point must rise with the support, not fall. If it falls, the sign of k in (4.3)



(or of `k_frt_ref`) is the place to change it.

2. **Resonator states.** The resonator has no anti-windup and no reset input: while the limiter is active or the gates are off, x_1 and x_2 keep integrating the error (Section 6.5). With the forward-backward step of rev. 03 the states no longer grow on their own, but a reset input (like the one of `frt_vde4110`) would still help after a long time with the gates off. The damping $\delta = 0.005$ keeps the gain of the model's resonator; $\delta = 0$ (zero steady-state error, poles on the circle) and $\delta = 0.015$ (the continuous design, gain 4.3, notch 33 dB) are the alternatives, both safe with this step.
3. **Islanding detector not wired.** In `afe_rst_ctrl.slx` the block exists in the library but is not connected: its inputs would be `ug_ab` of `grid_voltage_sequences`, the PLL angle, the limited pulsation and an enable (the settle time of the model), its output `island` a trip of the converter. With the thresholds of Table 2 it does not trip on the FRT test dip.
4. **Power-limitation release** on $|v_{\mathrm{neg}}| < 0.7$ is almost no condition: a tighter value (e.g. 0.05, with the same hysteresis idea as the positive sequence) would release the power limitation only when the unbalance is really gone.
5. `i_stack_ref` **in the model** is built with a Bus Creator; the block expects a 3×1 vector (a Mux).



12 Files

Table 4: Files of the package and of this note.

File	Content
folder `library/masked_models/afe_uvw_ctrl/` **of** `pwr-control/library`	
`afe_uvw_ctrl.slx`	the library with the five masked MATLAB Function blocks
`src/*.m`	the five functions, one file each, with their helper functions at the end (`moving_average_frac`, `pi_clamped`, `moving_sum_push`)
`afe_uvw_ctrl_param_example.m`	example parameter structure `p` with the numbers of the model and the observer gain formula (4.2)
`update_afe_uvw_ctrl.m`	reloads the sources into the blocks of the library (MATLAB)
`test/`	Simulink bench `test_afe_uvw_ctrl.slx`, its source, plant and parameters, the loop version and the 15 checks
`tools/make_afe_uvw_ctrl.py`, `load_sources_into_slx.py`, `check_afe_uvw_ctrl.py`	generator of the library and the bench without MATLAB; loader of the sources into the blocks of the MATLAB-saved library, without MATLAB (code-only changes); the check against the Python twin of the model's diagrams
`README.md`	description of the blocks, parameters, checks and use in the model
model `foundations_of_power_electronics/afe_uvw_three_phase/` **of** `pwr-control/modelization-and-control`	
`afe_rst_ctrl.slx`, `init_model.m`	the model with the blocks inside `afe_control`; the section "AFE control in uvw" of the init builds `param.afe.uvw`
this note, `repos_doc/afe_uvw_ctrl/` **of** `pwr-control/repositories-documentation`	
`afe_uvw_ctrl.tex`, `.pdf`	source and PDF
`afe_uvw_ctrl_figures.py`	the numbers of Section 6 and Figures 4–6 (numpy, matplotlib); writes `figures/afe_uvw_ctrl_numbers.txt`
`figures/`	the PDF figures of the script and the two PNG figures of the checks

References

[1] pwr-control, *library* repository, package `library/masked_models/afe_uvw_ctrl` (sources, README, tests), https://github.com/pwr-control/library.

[2] pwr-control, *modelization-and-control* repository, model `foundations_of_power_electronics/afe_uvw_three_phase`, https://github.com/pwr-control/modelization-and-control.

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[4] VDE-AR-N 4105:2018-11, *Generators connected to the low-voltage distribution network*, VDE FNN, 2018.

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